

Windows Authentication & Security Event Investigation

Project 2 — Cybersecurity Portfolio Incident Investigation Report

1. Executive Summary

This project documents a hands-on investigation of Windows Security Event Logs using Windows Event Viewer. The investigation examined authentication, privilege, account, and process-related events and included a controlled failed-login test. The available evidence did not indicate unauthorized access or malicious activity. The Event ID 4625 event was intentionally generated for investigative practice.

2. Objectives

Analyze Windows Security Event Logs; investigate successful and failed authentication; examine logon types, processes, and source addresses; review account and privilege activity; distinguish legitimate activity from potential security concerns; and document evidence using a SOC-style workflow.

3. Tools & Environment

Tool: Windows Event Viewer

Data Source: Windows Security Event Log

Environment: Personal Windows system

Investigation Type: Controlled educational/portfolio exercise

4. Event Review

Event ID	Description	Result
4624	Successful account logon	Normal interactive logon observed
4625	Failed account logon	Controlled failed-login event investigated
4648	Explicit credential use	Local activity involving svchost.exe observed
4672	Special privileges assigned	SYSTEM activity observed; likely normal
4688	New process created	No events available in examined log
4720	User account created	No events identified
4732	Member added to security-enabled local group	No events identified

5. Key Findings — Event ID 4624

Finding: A successful logon event with Logon Type 2 and the normal Windows user account was identified. Logon Type 2 represents an interactive logon, such as signing into Windows directly at the computer. **Assessment:** Likely benign and consistent with normal user activity.

6. Key Findings — Event ID 4648

Finding: An Event ID 4648 involved the normal Windows account, localhost, and svchost.exe. **Assessment:** Likely benign based on the available context. The event alone did not establish malicious activity.

7. Key Findings — Event ID 4672

Finding: An Event ID 4672 event associated with the SYSTEM account contained multiple special privileges.

Assessment: Likely normal Windows operating-system activity; no malicious behavior was established from the event.

8. Controlled Failed Authentication — Event ID 4625

A single incorrect password was intentionally entered at the Windows sign-in screen and the correct password was then used to sign in. This generated Event ID 4625 for controlled investigation purposes.

Field	Observed Value
Event ID	4625
Failure Reason	Unknown user name or bad password
Logon Type	2 — Interactive
Process	svchost.exe
Source Network Address	127.0.0.1

Assessment: Benign and controlled. 127.0.0.1 is the local loopback address, and the event was intentionally generated during the investigation. It should not be interpreted as evidence of a real-world compromise.

9. Account & Process Investigation

No Event ID 4720 events were identified, providing no evidence in the examined log of a newly created user account. No Event ID 4732 events were identified, providing no evidence of an account being added to a security-enabled local group. No Event ID 4688 events were available. This does not prove that no processes were created; it indicates that process-creation auditing data was not available in the portion of the Security log examined.

10. Security Analysis

The investigation did not identify evidence of unauthorized access, account creation, privilege escalation, or suspicious external authentication activity. The Event ID 4625 event was a controlled test. Analysts should correlate event type, account, logon type, process, source, timing, and known context before classifying an alert as malicious.

11. MITRE ATT&CK; Relevance

T1078 — Valid Accounts: Authentication events can be reviewed for potential misuse of legitimate credentials.

T1110 — Brute Force: Repeated failed authentication events can be investigated for possible password attacks.

T1098 — Account Manipulation: Account and group membership events can help identify unauthorized account or privilege changes.

No malicious execution of these techniques was identified during this controlled investigation.

12. Skills Demonstrated

Windows Event Viewer; Windows Security Event Logs; Event ID analysis; authentication log analysis; Logon Type interpretation; failed authentication investigation; account activity monitoring; privilege-event analysis; evidence collection; security event correlation; false-positive identification; incident documentation.

13. Conclusion

This investigation provided hands-on experience collecting and interpreting Windows security evidence. Normal Windows authentication and privilege events were reviewed, and a controlled failed-authentication event was generated and analyzed. The available evidence did not indicate unauthorized access or malicious activity.

14. Portfolio Disclaimer

This project was conducted on a personal Windows system for educational and portfolio purposes. The failed authentication event documented in this report was intentionally generated as part of the investigation and is not evidence of a real-world compromise.